

ABDELGHAFOUR MOUMAD

FUTUR INGÉNIEUR

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RESUME

Master’s student in Smart Industry with a strong drive to transform traditional manufacturing through digitalization and intelligent technologies. Focused on integrating robotics, IoT, and AI to enhance industrial efficiency, sustainability, and decision-making. Experienced in leading innovative projects, coordinating multidisciplinary teams, and developing digital solutions that bridge the gap between industrial operations and smart automation. Dedicated to shaping the future of Industry 4.0 through innovation, collaboration, and data-driven transformation

EDUCATION

Master’s Degree in Smart Industry	10/2024-
Université Sidi Mohamed Ben Abdellah faculté des Sciences Dhar El Mahraz	
Bachelor’s Degree in Physics – Electronics	9/2021 - 7/2024
Université hassan 2 faculté des sciences ben m'sik	2020 - 2021
High School Diploma – Experimental Sciences (Physics & Chemistry)	

COMPETENCES

Technical Skills

- **Automation & Industrial Maintenance**
- **Embedded Systems** : Arduino, ESP, Raspberry Pi, STM32, FPGA,NVIDIA NANO JETSON
- **Robotics Programming**: MELFA BASIC IV & V
- **Programming Languages & Low-Level Programming**: C, Python, MATLAB(SIMULINK), MicroC, Assembly
- **PLC Programming: Siemens (LAD, STL)**, Codesys , Tia portal
- **Containerization & Deployment**: Docker
- **3D Modeling & Simulation**: Fusion 360, SolidWorks, Unity 3D, Blender,catia v5 , easyeda
- **Advanced Technologies**: Digital Twins, Immersive Technologies, Cloud Computing, Blockchain, Hyperledge, Argus.
- **IoT & Wireless Sensor Networks (WSN)**
- **Industrial Digitalization & Smart Manufacturing**:
 - Industrial IoT Platforms**: Siemens MindSphere, ThingWorx, Azure IoT Hub
 - SCADA & HMI Systems**: WinCC, Ignition, TIA Portal Visualization
 - Industrial Communication Protocols**:Modbus, OPC-UA, MQTT, CAN Bus
 - MES & Digital Factory Tools**:Siemens Opcenter, Factory I/O, FlexSim
- **Data & AI for Industry**:
 - Data Analysis & Visualization**:Pandas, NumPy, Power BI, Grafana,matlab
 - Edge AI & Industrial Analytics**:TensorRT, ONNX Runtime, Edge Impulse
 - Predictive Maintenance & Anomaly Detection**: Time Series Modeling, AIoT
- **Software Engineering & System Integration**
 - Version Control**: Git, GitHub, GitLab
- **Emerging & Smart Tech**
 - AR/VR for Industry**: Unity XR
 - Cobotics & Human–Robot Interaction**: coppeliasim , cosimir , ROS2 HUMBLE, MoveIt,Gazeboo,rviz2
 - Digital Manufacturing Tools**: Additive Manufacturing (3D printing, slicers)

SOFT SKILLS

- Complex problem-solving
- Teamwork & interdisciplinary collaboration
- Leadership & team management
- Autonomy & initiative
- Agile project management
- Adaptability to new technologies
- Critical & analytical thinking
- Creativity & innovation
- Results-oriented mindset

INFORMATIONS COMPLEMENTAIRES

- **Languages:** English, French, arabic,amazigh
 - **Certifications:** NRC (uM6p), ARB (ENSAM), NURC (UPF) , Enactus(FSDM),Coursera(Python for Data Science, AI & Development-Supervised Machine Learning: Regression and Classification),Catia V5 Beginner to Advanced - Automotive and Industrial,SOLIDWORKS Foundations: Sketching and Extrusion,Modern Robotics, Course 1: Foundations of Robot Motion,Python for Data Science, AI & Development ,Learning Complete PCB Design: From an Idea to a Product
 - **Interests:** Intelligent robotics, AI, computer vision, digital twins, digitalization , industry 4.0
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ENGINEERING & R&D PROJECTS

- **Intelligent Quadcopter Drone** – Autonomous drone with GPS-based navigation (Raspberry Pi, LoRa, ESP32, PID).
- **Smart Cane for the Visually Impaired** – Assistive cane (RPI, GPS, ultrasonic sensors, camera, TTS) for urban guidance.
- **Quadruped Robot – NRC UM6P** – 4-legged wooden robot built in 24h for maze navigation & rescue missions.
- **Autonomous Waste Collector Robot** – Rover equipped with AI (RPI + camera + robotic arm) to detect & collect waste.
- **Smart Humidity Collection Kiosk** – Autonomous kiosk extracting humidity from air into potable water.
- **Parent-Child Safety Bracelet** – Connected wearable for child tracking with alert system.
- **Dual-Axis Smart Solar Panel** – Automatic solar tracking system with LDR sensors & servos for optimal energy capture.
- **SmartSweep – Autonomous Night Street Cleaner** – LiDAR, infrared cameras, GPS, and AI for autonomous cleaning.
- **ROBOT GESTION STOCK**-LIDAR-RPI 5-IMU -ENCODER

EXTRACURRICULAR ACTIVITIES

Project Manager – Robotics Club FSDM (Fès)

Coordination & Development of Technical Activities

- **Management of Multiple Projects:** As Project Manager, I supervised several technical and event-based initiatives simultaneously within the Robotics Club. This role required cross-functional coordination between specialized teams, including technical, communication, and logistics departments. I was responsible for planning and tracking project milestones, managing resources effectively, and ensuring deadlines were met.

- **Organization of Major Events:** I fully coordinated the Final-Year Project Day(PFE), from the planning stage to final execution, supporting students in structuring and presenting their projects. In parallel, I managed the club's complete participation in the Scientific Heritage Festival, including the preparation of technical demonstrations and showcasing innovative achievements developed by the club.

Right now, working on organizing a week-long about Innovation and Robotics Workshop and Robotics Atelier.

- **National Robotics Competitions – Project Manager:** I led the strategic participation of the Robotics Club FSDM in several high-level national competitions, simultaneously managing the technical, logistical, and human aspects of each project.

Directe a participating team:

- **NRC** – National Robotics Competition (UMGP, Benguerir)
- **NURC** – National University Robotics Challenge (UPF, Fès)
- **ARB** – African Robotic Brains (ENSAM, Meknès)

INTERNSHIP EXPERIENCE

Selecma Panel — Maintenance Technician (June 2024 – September 2024)

- Worked on industrial automation systems based on Siemens PLCs.
- Implemented and troubleshooted communication using the Profibus protocol.
- Participated in maintenance and programming tasks for automated control panels.

InnovLab Technology — Robotics Engineering Intern (July 2025 – October 2025)

- Worked on a logistics robot project for stock management.
- Used ROS2 Humble, SLAM, and Nav2 for navigation and mapping.
- Focused on software development and system integration (project still in progress, without physical realization).